

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – 10

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Class:TYIT	Batch:1
Date of Assignment:04/09/2026	Date/Time of Submission:04/09/2026

Data Preparation and Transformation using Tableau Prep

A) Connect and load data into Tableau Prep

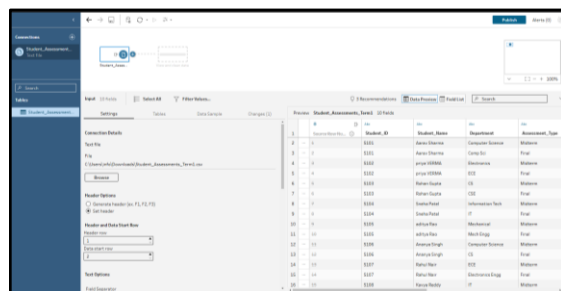
Step 1: Open Tableau Prep Builder

1. Launch Tableau Prep Builder on your machine.
2. On the start screen, locate the Connections pane on the left-hand side.

Step 2: Load the First Dataset

(Student_Assessments_Term1.csv)

1. In the Connections pane, click the + (Add connection) button.
2. In the list of file formats, select Text file.
3. Browse to the folder where you saved the CSVs, select Student_Assessments_Term1.csv, and click Open.
4. An Input Step icon named Student_Assessments_Term1 will appear on the Flow canvas.
5. canvas.



5. In the bottom Input configuration pane, verify:

- o Field Separator: Comma (,)
- o First row contains header: Checked

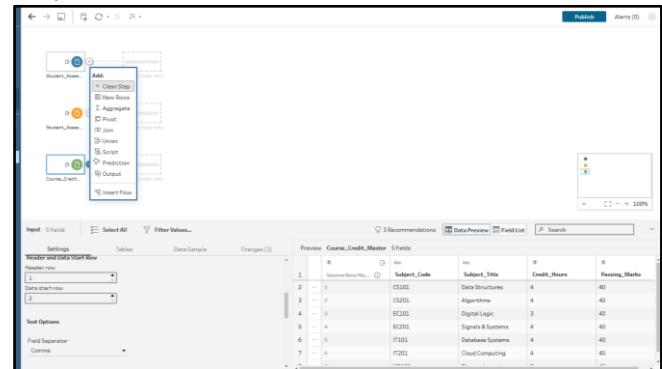
Step 3: Load the Second and Third Datasets

1. In the top-left Connections pane, click the + (Add connection) button again.
2. Select Text file, browse and select Student_Assessments_Term2.csv, and click Open.
3. Click + (Add connection) once more -> select Text file -> choose Course_Credit_Master.csv -> click Open.

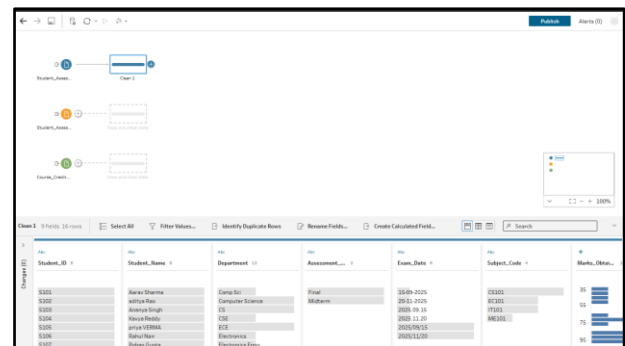
4. You will now see three separate Input icons on your Flow canvas.

Step 4: Add an Initial Clean Step to Term 1

1. On the Flow canvas, hover your mouse over the Student_Assessments_Term1 input icon.
2. Click the small + icon that appears to its right.
3. Select Clean Step (the step with the cleaning tool icon).



4. The Profile Pane will open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1.



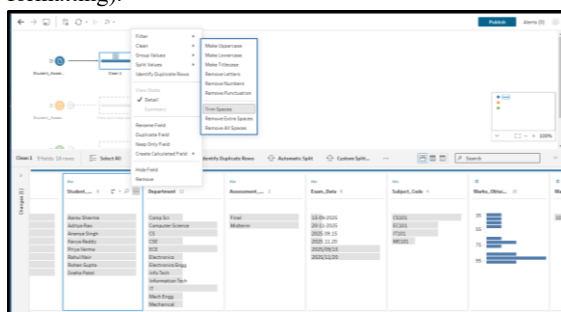
B) Clean and profile data

Step 1: Inspect the Profile Pane

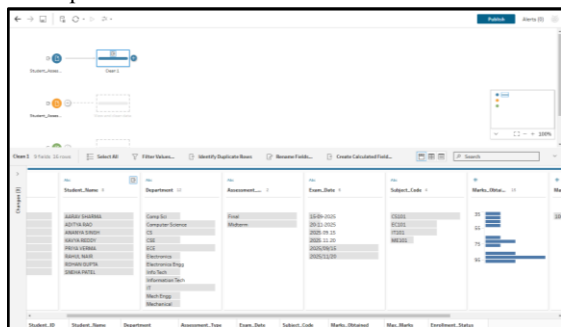
1. Click on the Clean 1 step connected to Student_Assessments_Term1.
2. Look at the cards displayed in the Profile Pane:
 - o Each card represents one column.
 - o Notice that Student_Name has multiple representations of the same student due to extra spaces and mixed casing (e.g., 'Aarav Sharma' vs 'Aarav Sharma', 'priya VERMA', 'aditya Rao').
 - o Notice that Exam_Date is recognized as a string (Abc) due to mixed separators (/, -, .).

Step 2: Clean Whitespace and Standardize Letter Casing

1. On the Student_Name profile card:
 - o Click the More options (...) icon at the top right of the card.
 - o Go to Clean --> select Trim Spaces (this strips all leading and trailing whitespace).
 - o Click More options (...) again --> Clean --> select Make Uppercase (or Make Lowercase / standard proper formatting).

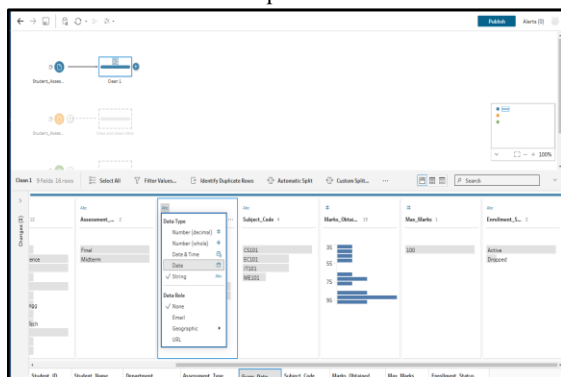


2. The card updates immediately, consolidating duplicate whitespace variations.

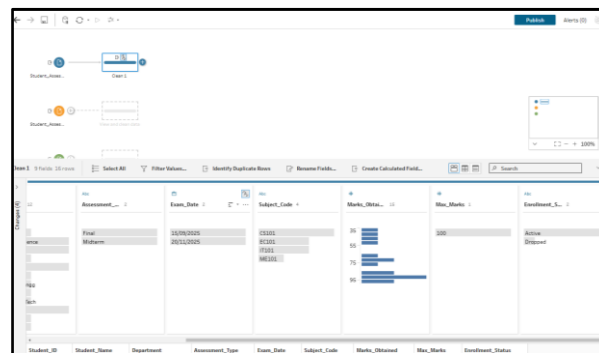


Step 3: Convert Exam_Date to Date Data Type

1. Locate the Exam_Date profile card.
2. Click the data type icon (currently showing Abc) directly above the column name.
3. Select Date from the dropdown menu.



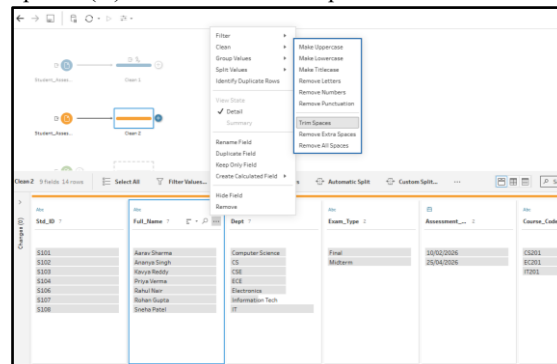
4. Tableau Prep automatically parses the various date formats (2025/09/15, 15-09-2025, 2025.11.20) into standardized calendar dates.



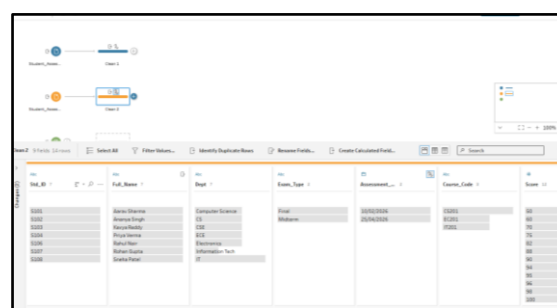
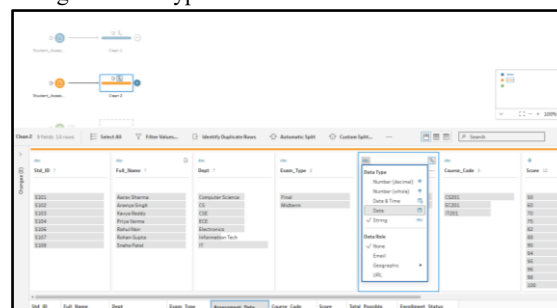
Step 4: Clean the Second Dataset

(Student_Assessments_Term2.csv)

1. On the Flow canvas, hover over the Student_Assessments_Term2 input icon.
2. Click + --> select Clean Step (creates Clean 2).
3. On the Full_Name card in Clean 2:
 - o Click More options (...) --> Clean --> Trim Spaces.



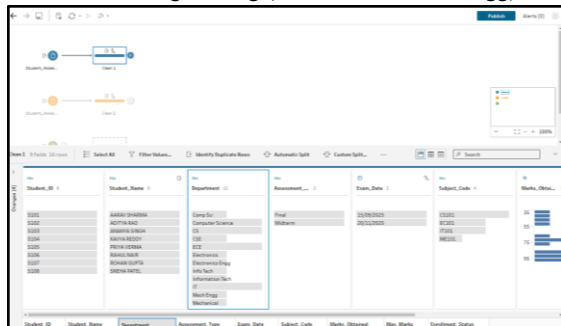
4. On the Assessment_Date card: Click the Abc icon --> change the data type to Date.



CJ Group and standardize values

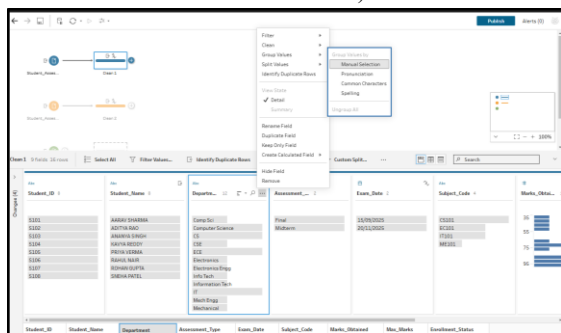
Step 1: Inspect the Department Field in Term 1 (Clean 1)

1. Click on the Clean 1 step connected to Student_Assessments_Term1.
2. Locate the Department card in the Profile Pane.
3. Observe that there are 12 distinct entries representing only 4 actual academic departments:
 - o Computer Science (Computer Science, Comp Sci, CS, CSE)
 - o Electronics (Electronics, ECE, Electronics Engg)
 - o Information Technology (Information Tech, IT, Info Tech)
 - o Mechanical Engineering (Mechanical, Mech Engg)

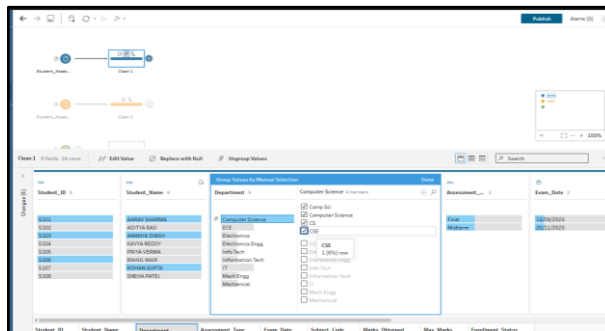


Step 2: Group Values Using Pronunciation / Spelling or Manual Selection

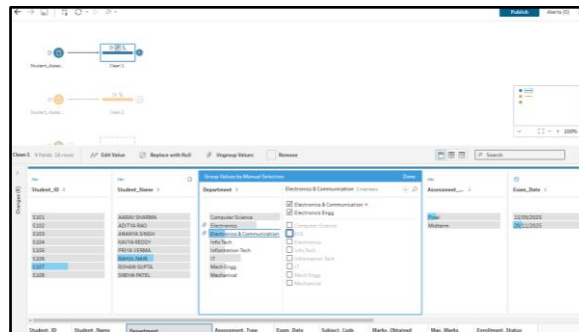
1. On the Department card, click the More options (...) menu at the top right.
2. Go to Group Values --> select Manual Selection (or Pronunciation / Common Characters).



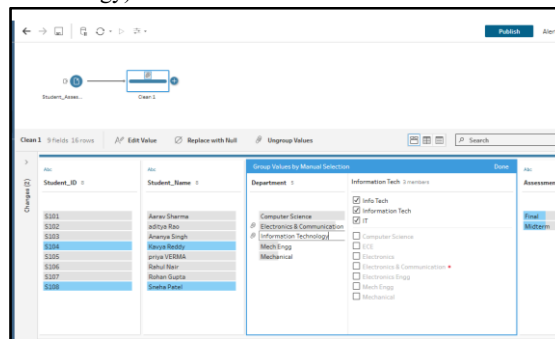
3. In the Grouping editor:
 - o Click on Computer Science in the left column. In the right-hand panel, check the boxes for Comp Sci, CS, and CSE.



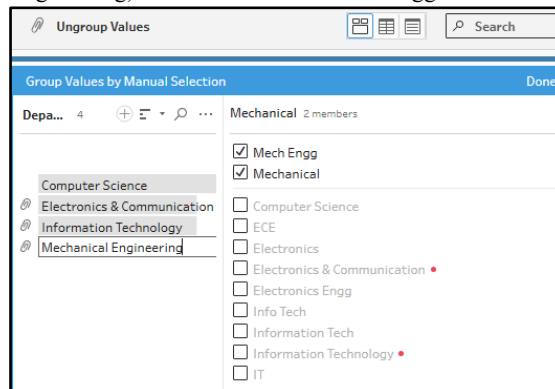
- o Click on Electronics (or rename the master value to Electronics & Communication). Check the boxes for ECE and Electronics Engg.



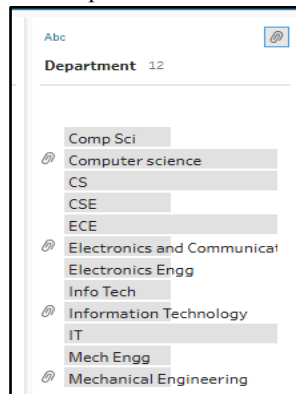
- o Click on Information Tech (or rename to Information Technology). Check the boxes for IT and Info Tech.

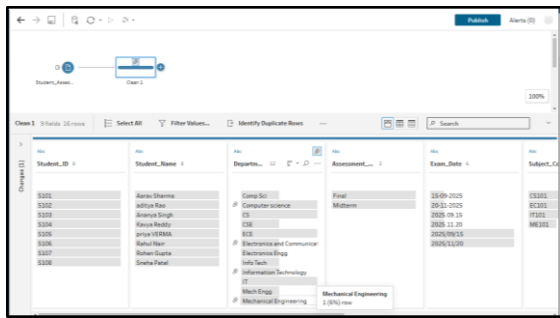


- o Click on Mechanical (or rename to Mechanical Engineering). Check the box for Mech Engg.



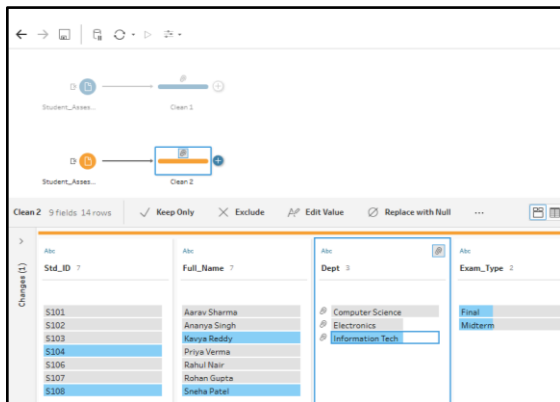
4. Click Done at the top right of the editor.
5. Notice that the paperclip icon appears next to grouped values, consolidating the 12 messy entries into exactly 4 clean departments.





Step 3: Repeat Grouping on Term 2 (Clean 2)

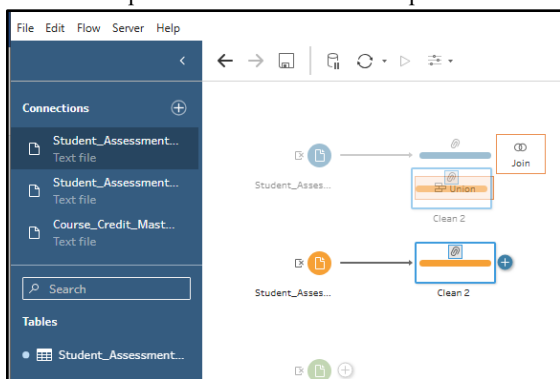
1. Click on the Clean 2 step connected to Student_Assessments_Term2.
2. Locate the Dept card.
3. Click More options (...) -->Group Values -->Manual Selection:
 - o Group CS, CSE, and Computer Science into Computer Science.
 - o Group ECE and Electronics into Electronics.
 - o Group IT and Information Tech into Information Tech.
4. Click Done.



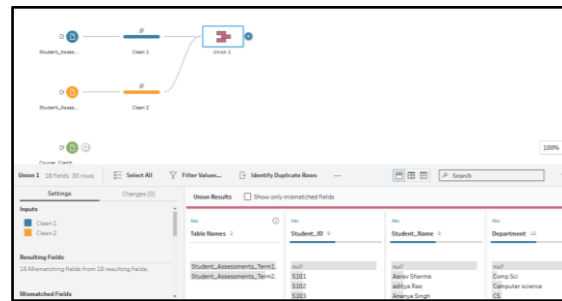
DJ Merge mismatched fields

Step 1: Create the Union Step on the Canvas

1. On the Flow canvas, click and drag the Clean 2 step icon over to the Clean 1 step icon.

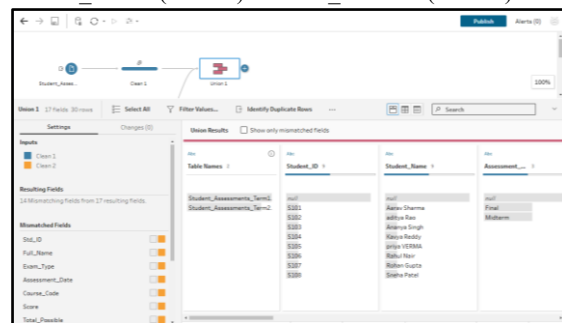


2. When the drop zone appears, hover over Union and release the mouse.
3. A new Union 1 step is created, connecting both transformation branches.



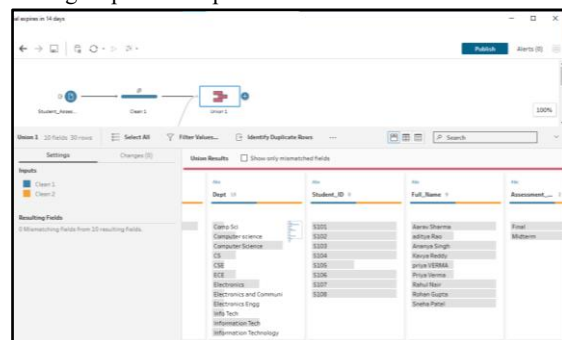
Step 2: Inspect Mismatched Fields in the Union Pane

1. Click on the Union 1 step.
2. In the bottom Union summary pane, look at the Mismatched Fields section.
3. Notice that because the two files used different naming conventions, Tableau Prep lists non-matching pairs (indicated by yellow and blue bars), such as:
 - o Student_ID (Term 1) vs Std_ID (Term 2)
 - o Student_Name (Term 1) vs Full_Name (Term 2)
 - o Department (Term 1) vs Dept (Term 2)
 - o Assessment_Type (Term 1) vs Exam_Type (Term 2)
 - o Exam_Date (Term 1) vs Assessment_Date (Term 2)
 - o Subject_Code (Term 1) vs Course_Code (Term 2)
 - o Marks_Obtained (Term 1) vs Score (Term 2)
 - o Max_Marks (Term 1) vs Total_Possible (Term 2)



Step 3: Merge the Mismatched Fields

1. In the Mismatched Fields list (or the union profile pane):
 - o Click and drag Std_ID directly on top of Student_ID to merge them into one column.
 - o Drag Full_Name onto Student_Name.
 - o Drag Dept onto Department.

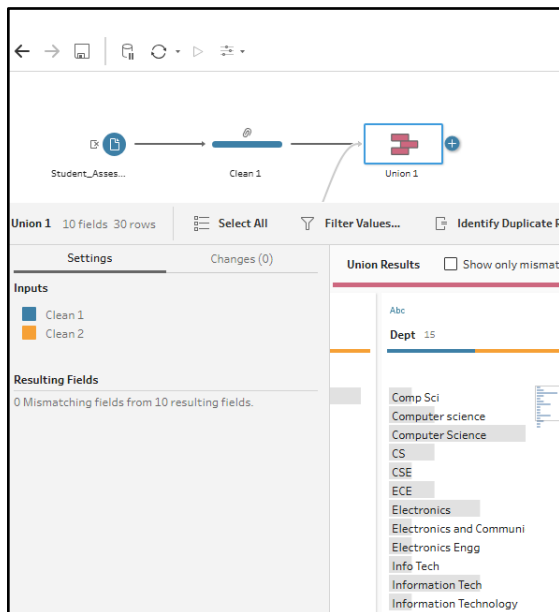


Note: If You cant drag it Observe in the left pane click the dept column the hover near department column's plus icon and add the dept column into it

- o Drag Exam_Type onto Assessment_Type.
- o Drag Assessment_Date onto Exam_Date.

- o Drag Course_Code onto Subject_Code.
- o Drag Score onto Marks_Obtained.
- o Drag Total_Possible onto Max_Marks.

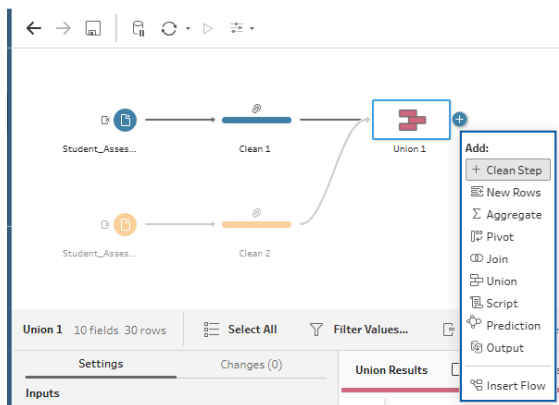
2. Notice that as you merge them, the mismatched count drops to 0, and each unified column displays both source colors.



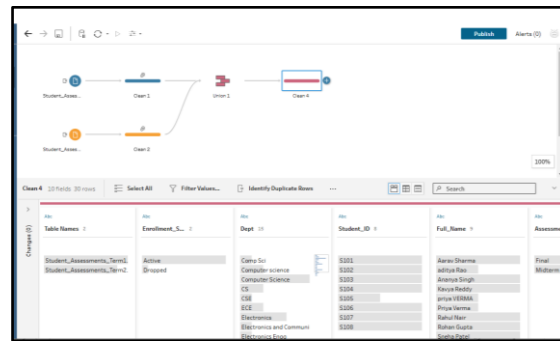
Step 4: Add a Clean Step Post-Union

1. Click the + icon next to the Union 1 node.

Trial expires in 14 days



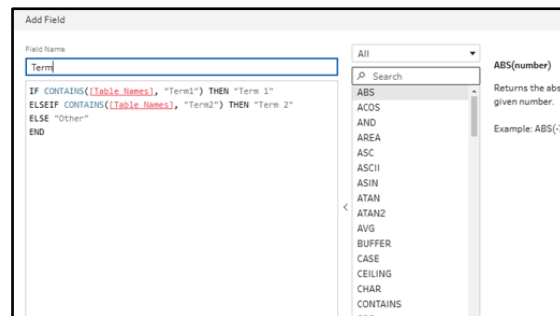
2. Select Clean Step (creates Clean 3). In Screenshot it is showing Clean 4 because I have created clean step for Course_Credit_Master.csv too.
3. Verify that the combined dataset now contains 30 total rows across 10 unified columns (including Table Names).



E] Perform aggregations in Prep

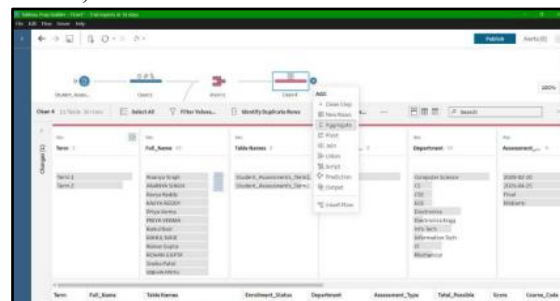
Step 1: Add a Term Identifier Field in Clean 3

1. Click on the Clean 3 step (the clean step created after Union 1).
2. In the toolbar above the Profile cards, click Create Calculated Field. You have to click three dots on the toolbar.
3. Set the Field Name to: Term
4. Enter the formula:
Code snippet
IF CONTAINS([Table Names], "Term1") THEN "Term 1"
ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"
ELSE "Other"
END
4. Click Save.

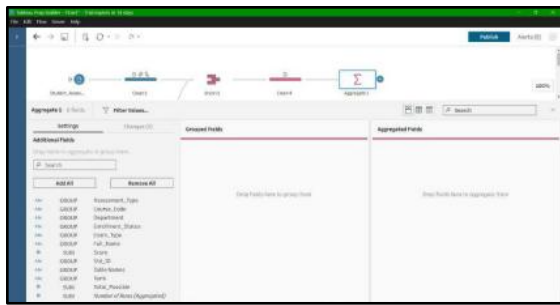


Step 2: Add an Aggregate Step to the Flow Canvas

1. Hover over the Clean 3 node on the Flow canvas.
2. Click the + icon next to it.
3. Select Aggregate (represented by the summation icon \$sum\$).



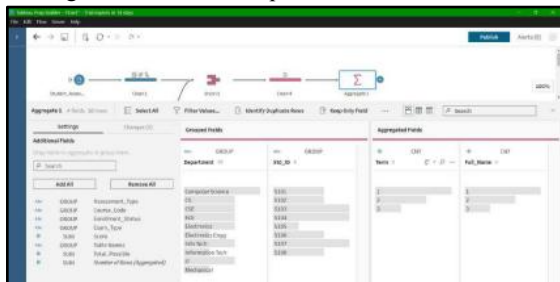
4. An Aggregate 1 step will appear, showing two drop zones in the configuration pane:
 - o Grouped Fields (Left Zone)
 - o Aggregated Fields (Right Zone)



Step 3: Configure Grouped Fields (Dimensions)

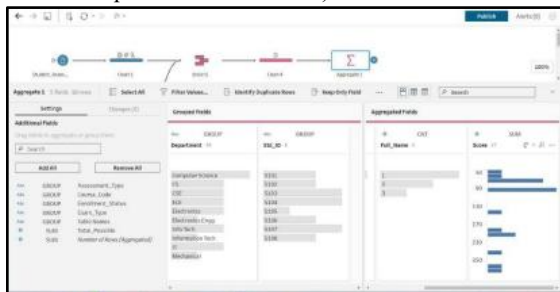
From the left-hand column list in the Aggregate step pane Your Column names could be different based on how you did previous steps of Union:

1. Drag Student_ID into the Grouped Fields box.
2. Drag Student_Name into the Grouped Fields box.
3. Drag Department into the Grouped Fields box.
4. Drag Term into the Grouped Fields box.

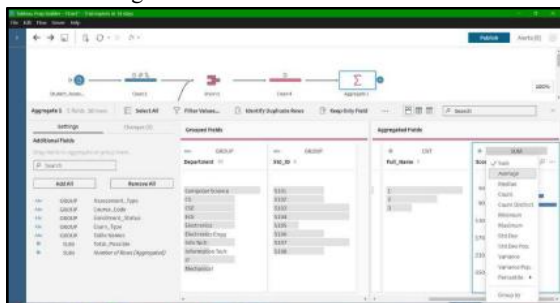


Step 4: Configure Aggregated Fields (Measures & Metrics)

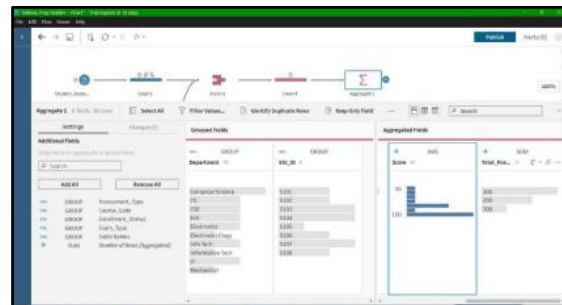
1. Total Marks: Drag Marks_Obtained into the Aggregated Fields box. (Ensure aggregation operation is set to Sum).



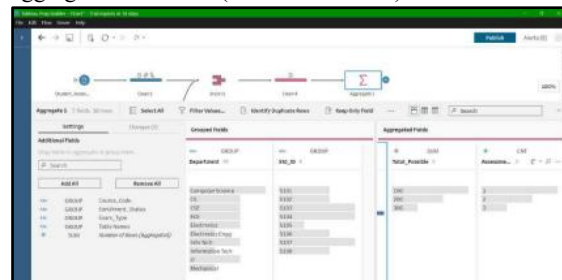
2. Average Score: Drag Marks_Obtained into the Aggregated Fields box a second time --> click on Sum --> change it to Average. There will be no field for dragging by any chance just change SUM to AVERAGE of the existing one.



2. Total Possible Marks: Drag Max_Marks into the Aggregated Fields box --> keep as Sum.



4. Assessment Count: Drag Assessment_Type (or Subject_Code) into the Aggregated Fields box --> set aggregation to Count (or Count Distinct).



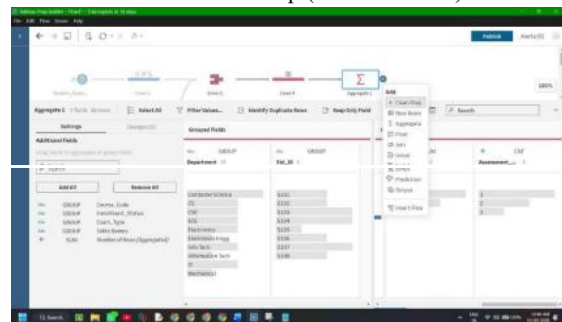
Step 5: Rename Aggregated Output Fields

In the Aggregated Fields panel, click on each field name to rename them cleanly:

- ☐ Marks_Obtained (Sum) --> Total_Marks_Obtained
- ☐ Marks_Obtained (Average) --> Average_Score
- ☐ Max_Marks (Sum) --> Total_Max_Marks
- ☐ Assessment_Type (Count) --> Assessments_Taken

Step 6: Add a Clean Step to Inspect Aggregated Output

1. Click the + icon next to the Aggregate 1 node -> select Clean Step (creates Clean4).

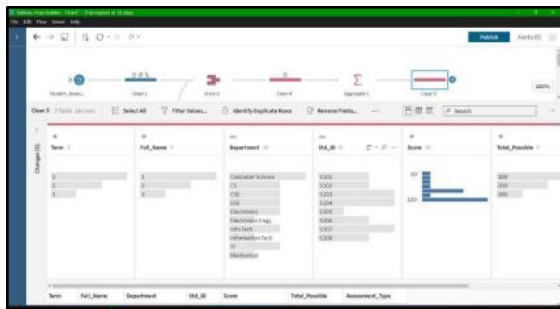


2. Look at the Profile Pane:
 - o The row count collapses from 30 individual assessment rows into 15 summary records or more (one row per student per term).

F | Apply Filters and Transformations

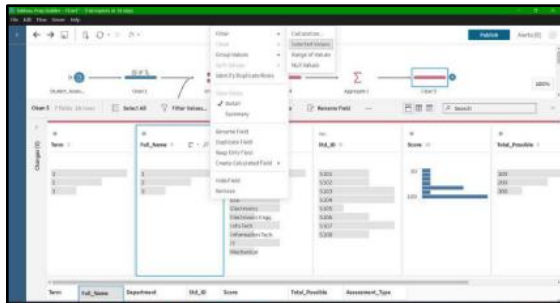
Step 1: Open the Transformation Step (Clean 4)

1. Click on the Clean 4 step connected after your Aggregate 1 step on the Flow canvas.
2. The Profile Pane below will show your aggregated columns: Student_ID, Student_Name, Department, Term, Total_Marks_Obtained, Average_Score, Total_Max_Marks, and Assessments_Taken. (Naming Will be different and completely depends on how you performed the Union Step)

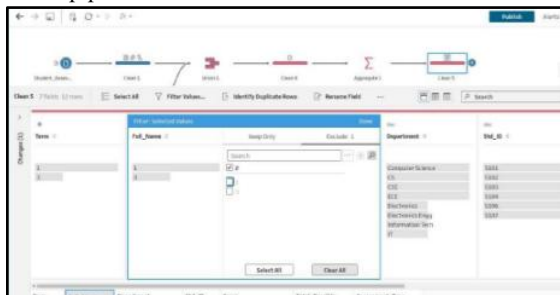


Step 2: Filter Out Inactive / Dropped Student Records

1. If Enrollment_Status is present on a profile card: o Click on the card --> click the value Dropped --> right-click and select Exclude (or click the filter icon at the top of the card and check only Active).
2. (Alternative via Filter calculation): o In the top toolbar, click Filter Values --> Selected Values....



- o Exclude Dropped so only Active students remain in the active pipeline.



Step 3: Create the Normalized Term Percentage Metric

1. In the toolbar above the Profile cards, click Create Calculated Field.
2. Field Name: Term_Percentage
3. Tableau Prep Formula:
Code snippet
$$\text{ROUND}([Total_Marks_Obtained] / [Total_Max_Marks]) * 100, 2)$$
$$\text{ROUND}([Score] / [Total_Possible]) * 100, 2)$$
 {This formula can be used if you couldn't able to rename your aggregate fields.}
4. Click Save.

Step 4: Create the Categorical Letter Grade Transformation

1. In the toolbar, click Create Calculated Field.
2. Field Name: Letter_Grade
3. Tableau Prep Formula:
Code snippet
IF [Term_Percentage] >= 85 THEN "A (Distinction)"

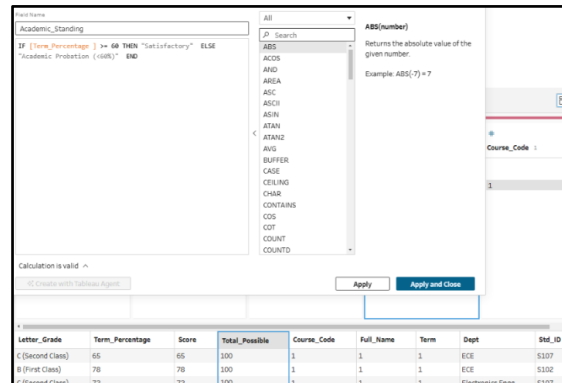
ELSEIF [Term_Percentage] >= 75 THEN "B (First Class)"
ELSEIF [Term_Percentage] >= 60 THEN "C (Second Class)"
ELSEIF [Term_Percentage] >= 40 THEN "D (Pass Class)"
ELSE "F (Fail / Re-appear)"
END

3. Click Save.



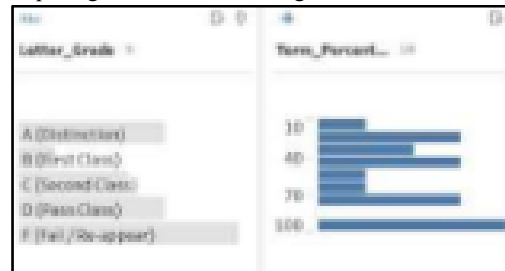
Step 5: Create Academic Standing Compliance Flag

1. Click Create Calculated Field.
2. Field Name: Academic_Standing
3. Tableau Prep Formula:
Code snippet
IF [Term_Percentage] >= 60 THEN "Satisfactory"
ELSE "Academic Probation (<60%)"
END
4. Click Save.



Step 6: Verify Transformed Profile Distributions

1. Look at the newly generated profile cards in Clean 4:
o Term_Percentage: Displays a smooth numeric distribution between 0% and 100%.
o Letter_Grade: Displays discrete categorical buckets (A (Distinction), B (First Class), etc.).
o Academic_Standing: Highlights students requiring academic counseling.



G] Create and automate data preparation flows

Step 1: Add an Output Step to the End of the Pipeline

1. On the Flow canvas, locate the final cleaning step (Clean 4).
2. Hover over the Clean 4 icon and click the + icon.
3. Select Output (represented by the download/save arrow icon).
4. An Output step node will appear connected directly to Clean 4.

Step 2: Configure Output Settings in the Bottom Pane

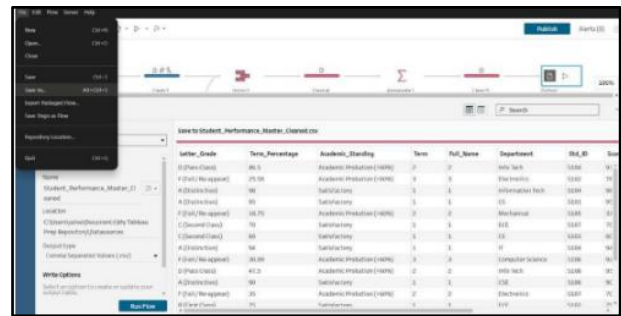
1. Click on the Output node to open its configuration panel below:
 - o Save output to: Select File (or Tableau Server / Tableau Cloud).
 - o Output type: Select Tableau Data Extract (.hyper) for high-speed dashboarding, or Comma Delimited Text (.csv) for universal export.
 - o Location: Click Browse and choose the directory where you want the final dataset saved.
 - o File Name: Enter Student_Performance_Master_Cleaned.
 - o Write Option: Select Create table (or Append to table if updating existing records).

Step 3: Run the Data Prep Flow

1. In the Output pane at the bottom-left, click the blue Run Flow button (or click the play icon  directly on the Output node on the canvas).
2. Tableau Prep will process all upstream nodes (Cleaning $\rightarrow$ Grouping $\rightarrow$ Unioning $\rightarrow$ Merging $\rightarrow$ Aggregating $\rightarrow$ Scoring).
3. A success notification will appear stating: "The flow ran successfully in X seconds."

Step 4: Save and Automate the Flow File

1. Go to the top menu bar $\rightarrow$ select File $\rightarrow$ Save As....
2. Name your workflow: Plaintext Student_Performance_Prep_Flow.tfl
3. (Automation Options in Production):
 - o Command Line Execution: You can automate this flow on Windows/Mac using the Tableau Prep Command Line Interface (CLI): `Bash tableau-prep-cli -t Student_Performance_Prep_Flow.tfl`
 - o Tableau Prep Conductor: If connected to Tableau Cloud/Server, you can publish the flow and set a scheduled refresh (e.g., every Monday at 6:00 AM) to automatically run whenever new semester files are added.



Letter_Grade	Term_Percentage	Academic_Standings	Term	Full_Name	Department	ID
A (Distinction)	95.00	Academic Production (100%)	1	John Smith	Information Tech	10001
A (Distinction)	90.00	Academic Production (100%)	2	Jane Doe	Information Tech	10002
A (Distinction)	85.00	Academic Production (100%)	3	John Smith	Information Tech	10003
A (Distinction)	80.00	Academic Production (100%)	4	Jane Doe	Information Tech	10004
A (Distinction)	75.00	Academic Production (100%)	5	John Smith	Information Tech	10005
A (Distinction)	70.00	Academic Production (100%)	6	Jane Doe	Information Tech	10006
A (Distinction)	65.00	Academic Production (100%)	7	John Smith	Information Tech	10007
A (Distinction)	60.00	Academic Production (100%)	8	Jane Doe	Information Tech	10008
A (Distinction)	55.00	Academic Production (100%)	9	John Smith	Information Tech	10009
A (Distinction)	50.00	Academic Production (100%)	10	Jane Doe	Information Tech	10010

Step 5: Verify the Final Master Dataset in Tableau Desktop

1. Open Tableau Desktop.
2. Connect to the exported Student_Performance_Master_Cleaned.csv (or .hyper).
3. Build the self-performance dashboard using the pre-calculated

